

AUNTOR CHAKMA

[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Email](#) | [Google Scholar](#)

RESEARCH INTEREST

Primary: Explainable AI (XAI), Generative AI, Computer Vision, Artificial Intelligence

Secondary: Agentic AI, Large Language Models, Human-AI Interaction, Data Science

PUBLICATION AND PREPRINT

- Sagar Mohajan, G.M. Sakhawat Hossain, Md Mynoddin, Rishita Chakma, Tanjim Mahmud, Punam Kanungoe, Auntor Chakma. (2026). "A Deep Learning-Based Framework for Bangla Vehicle License Plate Detection and Recognition." In: Palaiahnakote, S., et al. (eds) ICDSAIA 2025. Communications in Computer and Information Science, vol 2683. Springer, Cham, pp 220-235. [\[Paper Link\]](#)
- Auntor Chakma, et al. "Interpretable Diabetic Retinopathy Classification Using Diffusion-Based Synthetic Images and Explainable Ensemble Method." Manuscript in Preparation.
- Auntor Chakma, et al. "Attentive TabNet Framework for Multiclass Intrusion Detection in Internet of Medical Things." [Preprint]

EXPERIENCE

1. Research Intern

Jan 2026 – Present

ELITE Research Lab LLC, Queens, NY, USA (Remote)

Responsibilities:

- Awaiting project assignment to contribute to research initiatives in AI and machine learning
- Assisting with literature review and identifying research gaps in Generative AI and Computer Vision domains

2. Research Collaborator

Aug 2025 – Dec 2025

ELITE Research Lab LLC, Queens, NY, USA (Remote)

Research: Diabetic Retinopathy Classification using Generative AI and XAI

Responsibilities:

- Developed ensemble framework achieving 91.65% accuracy for seven-stage diabetic retinopathy classification
- Implemented Stable Diffusion 2.1 with ControlNet-Canny for addressing severe class imbalance
- Applied Grad-CAM++ explainability achieving 91.67% macro F1-score for clinical interpretability
- Combined top 3 CNN architectures improving baseline accuracy by 2.09%

3. Research Assistant

Dec 2024 – Jun 2025

Department of Computer Science and Engineering

Rangamati Science and Technology University, Bangladesh

Research: Machine Learning for Real-Time Malicious Network Traffic Detection

Personal Invigilator: Assistant Professor Md Mynoddin

Responsibilities:

- Collected and preprocessed real-time network traffic dataset from Bangladesh telecommunications
- Trained CNN-based deep learning models for malicious traffic detection
- Conducted literature review and identified research gaps for manuscript preparation

4. Software Engineer Intern

Mar 2024 – Jun 2024

Cellescope Limited

Dhaka, Bangladesh

Responsibilities:

- Assisted in backend systems using Java, Spring Boot, RESTful APIs, and MySQL
- Completed training in Agile methodologies (sprint planning, standups)
- Practiced Git version control workflows (branching, pull requests)

EDUCATION

Bachelor of Science in Computer Science and Engineering

Rangamati Science and Technology University, Bangladesh

Sep 2018 – Feb 2025

CGPA: 3.50/4.00

Thesis: Attentive TabNet Framework for Multiclass Intrusion Detection in IoMT.

Achieved 97.96% accuracy on 89 million IoMT network traffic records.

Higher Secondary Certificate (HSC)

Notre Dame College, Dhaka, Bangladesh

2015 – 2017

GPA: 5.00/5.00

SKILLS

- **Programming Languages:** Python, Java, C
- **ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, OpenCV, Hugging Face Transformers
- **Data Science:** NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering, Statistical Analysis
- **AI Techniques:** Stable Diffusion, ControlNet, GANs, Grad-CAM, Attention Mechanisms
- **Software Development:** Git, GitHub, RESTful APIs, Spring Boot, Django, Agile
- **Databases:** MySQL, MongoDB, Oracle, SQLite
- **Tools:** LaTeX, Jupyter Notebook, Linux, Wireshark, Microsoft Office

ACHIEVEMENTS AND AWARDS

- Merit Scholarship, Rangamati Science and Technology University (2023-2024, 2024-2025)
- Chittagong Hill Tracts Development Board Scholarship (2018, 2023, 2024, 2025)
- Rangamati Hill District Council Scholarship (2023, 2025)
- Talent Pool Scholarship in SSC and HSC by DSHE, Government of Bangladesh (2015, 2017)
- Third Place, Inter University Hackathon (2023)

SERVICES

- **AI Seminar Host**, Educational Campuses, Bangladesh (2023-2024). Hosted seminars on artificial intelligence and AI safety across educational campuses, educating students and faculty about AI applications, ethical considerations, and safety protocols.
- **Conference Website Developer**, RMSTU International Bioscience Conference (2024-2025). Built conference website (icbc.rmstu-conf.ac.bd) as part of 6-member development team. Worked as security analyst and backend developer.
- **Technical Session Coordinator**, RMSTU International Bioscience Conference, May 16-18, 2025. Coordinated technical sessions and assisted with conference logistics.
- **Winter Relief Distribution Volunteer**, Rangamati Hill District Rural Areas (Winter 2023, 2024). Distributed books and warm clothes to underprivileged communities.
- **Volunteer**, Bangladesh Red Crescent Society (2023-2024). Supported disaster response initiatives and community health awareness programs.